

Dashboard §11.4.170 host-render visual-proof increment — captured evidence

Revision: 1 **Last modified:** 2026-07-09T00:00:00Z **Run-id:** 20260709-dashboar-hostrender **Repo HEAD at capture:** de71a6343a89a20ba12fabedc54b0d1a2e0d7c7f (branch main) **Scope:** dashboard/ only (frontend). Server + clients//ota-manager/ untouched. **Constitution:** §11.4.170 (device-independent host-side rendered-pixel proof), §11.4.107(10) (self-validated golden-good/golden-bad analyzer), §11.4.5/§11.4.69 (captured evidence), §11.4.3 (honest SKIP), §11.4.6 (no guessing).

1. Environment

Item	Value
node	v24.18.0
go	go1.26.4
@playwright/test	1.60.0 (CLI 1.61.1)
chromium headless-shell build	1223 (downloaded via npx playwright install chromium)
added devDeps	pixelmatch@5.3.0, pngjs@7.0.0, @types/pngjs@6.0.5

node_modules was empty at start; installed via npm ci.

2. Existing suite result (baseline)

2.1 Component/unit (vitest) — `npm run test:run`

10 files, 93 tests, 93 passed. (React Router v7 future-flag warnings only; no failures.)

2.2 Playwright browser e2e (real SPA + real Go control plane) — `npx playwright test`

Full log: <logs/existing-e2e-baseline.log>

22 passed, 1 failed. Passing includes: `a11y.spec.ts` (axe WCAG2a/2aa/21a/21aa, Login + Overview, **0 critical/serious violations**), `smoke.spec.ts` (5), `deployments.spec.ts` (4), `releases.spec.ts` (2), `groups-detail.spec.ts` (4), `fleet-detail.spec.ts` (2), `populated-detail.spec.ts` (3).

The 1 failure is `record-all-screens.spec.ts` → “**record and verify all dashboard screens**” (the whole-suite screen *recorder*), which timed out waiting for the Overview `<h1>` after login. This is a **pre-existing** shared-in-memory-server ordering/flake in the recorder, **not introduced by this increment**: the byte-identical login → Overview assertion in `smoke.spec.ts:27` (“operator logs in and the Overview screen renders”) PASSED in the same run. Honestly reported per §11.4.6; out of scope for this increment (recorder spec, no source change made).

3. NEW §11.4.170 host-render increment

3.1 What was rendered, how

- **Screen: Login** (`src/screens/LoginScreen.tsx`) — the most self-contained screen (renders fully client-side; only calls `/api` on submit).
- **How:** real component pixels rendered ON THE HOST by headless Chromium against the vite-served SPA — **no device, no emulator, no running Go backend**. A dedicated config `dashboard/playwright.hostrender.config.ts` boots **only vite** (port 4318, `HELIX_API_TARGET=http://localhost:9` dummy) with **no globalSetup** — making the proof genuinely device- AND backend-independent.

- **Spec:** `dashboard/hostrender/login.hostrender.spec.ts`
- **Run:** `npm run e2e:hostrender (playwright test -- config=playwright.hostrender.config.ts)`
- **Result:** **4 passed** — log `logs/hostrender-run.log`.

3.2 Host-rendered PNG(s)

File	Dims	Meaning
<code>login-light-actual.png</code>	400×242	the Login card , real host pixels (LIGHT)
<code>login-light-viewport.png</code>	1280×720	full-viewport host render
<code>dashboard/hostrender/ login.hostrender.spec.ts- snapshots/login-card- light-chromium-linux.png</code>	400×242	committed golden baseline (byte- identical, 9193 B)

Machine-read (§11.4.107): `login-light-actual.png` shows the real component — title “Helix OTA — operator login”, “Username (email)” field with placeholder `operator@example.com`, “Password” field, blue “Sign in” button.

3.3 Dual oracle

(i) **Golden image-diff** — two independent, self-validated analyzers:

- *Playwright* `toHaveScreenshot()` against the committed baseline:
 - golden-good test **PASSES** (render matches baseline).
 - `toHaveScreenshot` self-validated test **PASSES** by proving the **committed baseline REJECTS** a mutated render (assertion throws → `.rejects.toThrow()`).
- *Explicit* pixelmatch *analyzer* — `image-diff-selfcheck.json`: |
 comparison | mismatched px | ratio | |—|—|—| | good ↔ good
 (identical reload) | **0** | 0.000 | | good ↔ mutated | **120160** | **0.1304**
 (**13.0%**) |

Verdict: **SELF-VALIDATED: analyzer passes golden-good AND flags golden-bad**. Diff artifacts: `diff-good.png`, `diff-bad.png`, `diff-good-vs-bad.png` — the diff image localizes the regression precisely over the card (red), margin unchanged (white).

(ii) OCR/text + DOM-bounds layout oracle — self-validated: - golden-good `layout-oracle-good.json`: `pass:true, failures:[]`. Asserts the **actually-rendered** text (`section.innerText`) contains all key labels (“Helix OTA — operator login”, “Username (email)”, “Password”, “Sign in”), every control box is on-screen / non-clipped / non-degenerate ($\geq 8 \times 8$), and **no control overlaps another**. - golden-bad `layout-oracle-bad.json`: `pass:false`. After injecting the canonical §11.4.170 regression (title `display:none` + “broken giant button” pinned over the form), the SAME oracle FLAGS it:

missing rendered label: “Helix OTA — operator login”, control “title” not rendered, and control **overlaps** — proving the oracle is not a rubber stamp.

Value/token-equality is NOT used as the proof (that would violate §11.4.170); the proof is rendered pixels + rendered text/bounds.

4. Honest gaps (§11.4.6)

- **DARK THEME NOT PROVEN — the dashboard ships LIGHT ONLY.** No dark mode exists in the code, so only the light `screen×state` is rendered here. The dark variant per `screen×state` is an explicit **TODO**: implement a dark theme, then add `login-card-dark` golden + dark oracle runs. This increment does **not** claim a dark variant exists.
 - **One screen only.** This is the FIRST §11.4.170 increment (Login). The other 7 screens (Overview, ArtifactUpload, Releases, Deployments, Fleet, Groups, Audit) remain to be host-render-proven per `screen×state` — follow-on work reusing this harness.
 - **True OCR (tesseract/vision) not wired.** The layout oracle reads the browser’s *rendered* text (`innerText`, which drops `display:none`) + DOM bounds rather than pixel OCR. This is the “text/DOM-bounds” arm of the §11.4.170 oracle; a pixel-OCR arm can be added later. The image-diff arm IS true rendered pixels.
 - **record-all-screens.spec.ts** pre-existing flake (see §2.2) is untouched.
-

5. Files written by this increment

Source / harness (in `dashboard/`, in-scope): - `dashboard/playwright.hostrender.config.ts` (new) - `dashboard/hostrender/login.hostrender.spec.ts` (new) - `dashboard/hostrender/`

login.hostrender.spec.ts-snapshots/login-card-light-chromium-linux.png (new committed golden) - dashboard/package.json (+e2e:hostrender script, +3 devDeps)

Evidence (docs/qa/20260709-dashboard-hostrender/): - EVIDENCE.md (this file) - login-light-actual.png, login-light-viewport.png - diff-good.png, diff-bad.png, diff-good-vs-bad.png - image-diff-selfcheck.json, layout-oracle-good.json, layout-oracle-bad.json - logs/existing-e2e-baseline.log, logs/hostrender-run.log

No git add/commit/push performed (per task constraints).